

Student Number: Teacher:

BAULKHAM HILLS HIGH SCHOOL

Assessment Task

December, 2010

Year 12

Mathematics Extension 2

Directions to candidates:

- Reading time: 5 minutes
- Working time: 50 minutes
- Attempt ALL questions
- Use black or blue pen only
- Start a new page for each question

Marks:

1	
2	
3	
4	
TOTAL /36	
Percentage	

Question 1 (9 marks) Start on a new page

a) If $z = 2 + 3i$ and $w = 1 - i$, express in the form $a + ib$

i) $\bar{z}$ 1

ii) $\frac{z}{w}$ 2

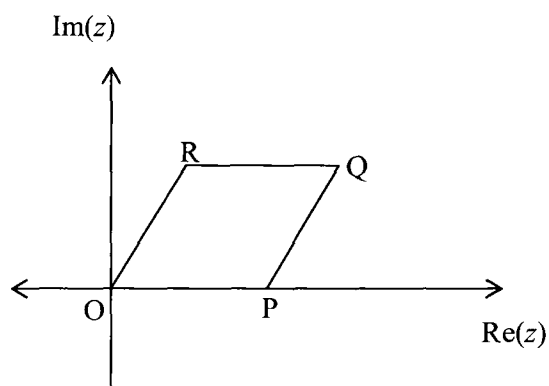
b) i) Find $\sqrt{6i-8}$ expressing your answers in the form $a + ib$ 2

ii) Hence solve $2z^2 - (3 + i)z + 2 = 0$ 2

c) Find the least value of $|z|$ when $|z - 4 - 3i| = 3$ 2

Question 2 (9 marks) Start on a new page

a) OPQR is a rhombus. O lies at the origin, P on the real axis and R corresponds to the complex number $1 + \sqrt{3}i$



i) Find the complex number corresponding to Q. 2

ii) If the figure is rotated anticlockwise by 60° about O to form a new rhombus $OP'Q'R'$, show this on an Argand Diagram and find the complex number corresponding to the vertex at Q'. 2

b) Draw neat sketches of the following

i) $\arg\left(\frac{z - 1 + i}{z + 2}\right) = -\frac{\pi}{3}$ 3

ii) $0 < \arg(1 - i)z < \frac{\pi}{6}$ 2

Question 3 (9 marks) Start on a new page

- a) i) Solve the equation $z^5 - 1 = 0$ over the complex number field. 1
- ii) Hence factorise $z^5 - 1$ in terms of real linear and quadratic factors. 2
- iii) Hence, show that $\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$ 3
- b) If $z = r(\cos \theta + i \sin \theta)$ show that $\frac{z}{z^2 + r^2}$ is purely real and find its value. 3

Question 4 (9 marks) Start on a new page

- a) If $1, w, w^2$ are the 3 cube roots of unity and given that $1+w+w^2=0$:
- i) Show that $1 + w$ is a root of the equation $z^3 - 3z^2 + 3z - 2 = 0$ 2
- ii) Find the integer root of this equation AND also the third root in terms of w 2
- b) i) Show that the locus of $\operatorname{Re}(z - 2) + 2 = |z - 2|$ is the parabola $y^2 = 4(x - 1)$. 2
- ii) Hence, show that when z satisfies the above relation, the principal argument of z lies between $-\frac{\pi}{4}$ and $\frac{\pi}{4}$ inclusive. 3

End of Paper

1a) i) $2-3i$

ii) $= \frac{2+3i}{1-i} \times \frac{1+i}{1+i}$
 $= \frac{-1}{2} + \frac{5i}{2}$

bi) $\sqrt{6i-8} = x+iy$ (check x, y are real)

$6i-8 = x^2 - y^2 + 2xyi$

Equating real & imaginary parts

$x^2 - y^2 = -8$ (1)

$2xy = 6$ (2)

from (2) $y = \frac{3}{x}$

sub in (1) $x^2 - \frac{9}{x^2} = -8$

$x^4 + 8x^2 - 9 = 0$

$(x^2 + 9)(x^2 - 1) = 0$

$x^2 + 9 = 0$

$x^2 - 1 = 0$

no real sol

$x = \pm 1$

$y = \pm 3$

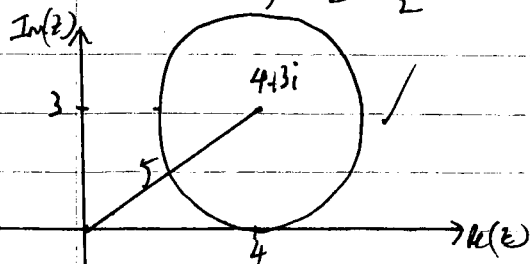
$\therefore \sqrt{6i-8} = \pm(1+3i)$

bi) $z = \frac{3+1i \pm \sqrt{8+6i-4 \times 2 \times 2}}{4}$

$= \frac{3+1i \pm \sqrt{6i-8}}{4}$

$= \frac{3+1i \pm (1+3i)}{4}$

$\therefore z = 1+i, \frac{1}{2} - \frac{i}{2}$



$\min |z| = 5-3 = 2$

2a) i) $|\vec{OP}| = |\vec{OR}| = \sqrt{1+3} = 2$

$\therefore P$ is $(2, 0)$

$\vec{OQ} = \vec{OP} + \vec{OR}$ ($\vec{OP} = \vec{OR}$)

$= 2 + 1 + \sqrt{3}i$

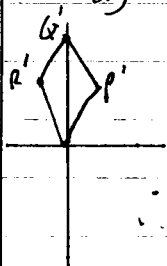
$\therefore Q = 3 + \sqrt{3}i$

ii) $\vec{OQ}' = \vec{OQ} \times \cos 60^\circ$

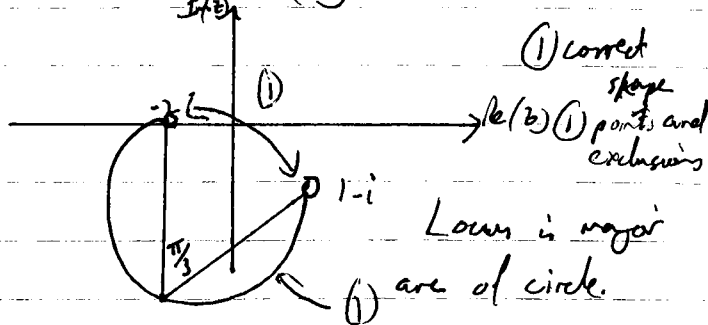
$= (3 + \sqrt{3}i) \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$

$= \left(\frac{3}{2} - \frac{3}{2} \right) + i \left(\frac{\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} \right)$

$\therefore Q$ is $(0, 2\sqrt{3}i)$
 or $\sqrt{3} \text{ cis } \frac{\pi}{2}$



b) i) $\arg \left(\frac{z - (1-i)}{z - (-2)} \right) = -\frac{\pi}{3}$

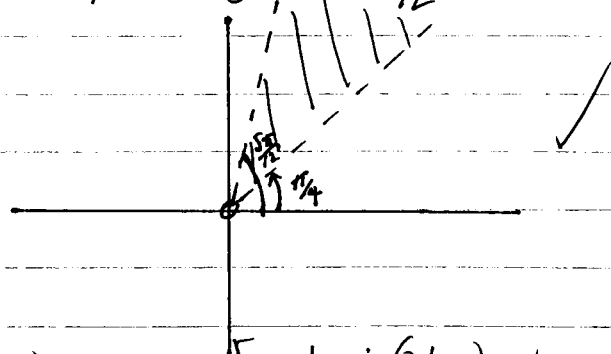


(ii) $0 < \arg(1-i)z < \frac{\pi}{6}$

$0 < \arg(1-i) + \arg(z) < \frac{\pi}{6}$

$0 < -\frac{\pi}{4} + \arg z < \frac{\pi}{6}$

$\frac{\pi}{4} < \arg z < \frac{5\pi}{12}$



3a) i) $z = 1 \text{ cis } (2k\pi)$ $k=0,1,2,3,4$

$\therefore z = \text{cis } \frac{2k\pi}{5}$ $k=0,1,2,3,4$

a) ii) $z_1 = 1$

$z_2 = \text{cis } \frac{2\pi}{5}$

$z_3 = \text{cis } \frac{4\pi}{5}$

$z_4 = \text{cis } \frac{6\pi}{5} = \bar{z}_3$

$$z^5 - 1 = (z-1)(z-\bar{z}_1)(z-\bar{z}_2)(z-\bar{z}_3)(z-\bar{z}_4)$$

$$= (z-1)(z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1)(z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2)$$

$$z^5 - 1 = (z-1)(z^2 - 2\cos\frac{2\pi}{5}z + 1)(z^2 - 2\cos\frac{4\pi}{5}z + 1)$$

iii) since $z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$

$$1z^4 + z^3 + z^2 + z + 1 = (z^2 - 2\cos\frac{2\pi}{5}z + 1)(z^2 - 2\cos\frac{4\pi}{5}z + 1)$$

Equating coeff of z^2 :

$$1 = 1 + 4\cos\frac{2\pi}{5}\cos\frac{4\pi}{5} + 1$$

$$4\cos\frac{2\pi}{5}\cos\frac{4\pi}{5} = -1$$

$$\cos\frac{2\pi}{5}\cos\frac{4\pi}{5} = -\frac{1}{4}$$

But $\cos\frac{4\pi}{5} = -\cos\frac{\pi}{5}$

$$\therefore \cos\frac{2\pi}{5} \times (-\cos\frac{\pi}{5}) = -\frac{1}{4}$$

$$\cos\frac{\pi}{5}\cos\frac{2\pi}{5} = \frac{1}{4} \text{ as req'd.}$$

3b) $\frac{z}{z^2 + r^2} = \frac{r(\cos\theta + i\sin\theta)}{r^2(\cos 2\theta + i\sin 2\theta + 1)}$

$$= \frac{\cos\theta + i\sin\theta}{r(2\cos^2\theta + 2i\sin\theta\cos\theta)}$$

$$= \frac{\cos\theta + i\sin\theta}{2r\cos\theta(\cos\theta + i\sin\theta)}$$

$$= \frac{1}{2r\cos\theta} \text{ or } \frac{1}{2r}\sec\theta$$

4a) i) $1 + w = -w^2$ since $1 + w + w^2 = 0$

$$P(-w^2) = (-w^2)^3 - 3(-w^2)^2 + 3(-w^2) - 2$$

$$= -(w^3)^2 - 3w^4 - 3w^2 - 2$$

$$= -3(1 + w^3 \cdot w + w^2)$$

$$= -3(1 + w + w^2)$$

$$= -3 \times 0$$

$$= 0$$

$\therefore 1 + w(-w^2)$ is a root.

ii) $P(z) = 8 - 12z + 6z^2 - z^3 = 0$

$\therefore z$ is an integer root.

Sum of roots one at a time (where α is 3rd root)

$$2 + (1 + w) + \alpha = 3$$

$$\alpha = -w$$

$\therefore$ Third root is $-w$.

4b) i) let $z = x + iy$

$$\operatorname{Re}(z - 2) + 2 = |z - 2|$$

$$x - 2 + 2 = |x - 2 + iy|$$

$$x = \sqrt{(x-2)^2 + y^2}$$

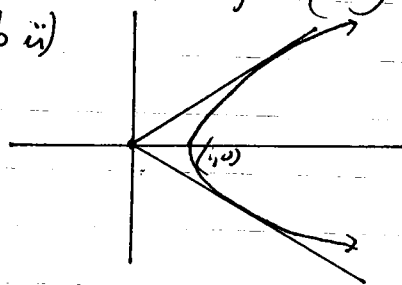
$$x^2 = x^2 - 4x + 4 + y^2$$

$$y^2 = 4x - 4$$

$$y^2 = 4(x-1)$$

$\therefore$ Parabola $y^2 = 4(x-1)$ is locus.

b ii)



let $y = mx$. Max and min arguments will be when $y = mx$ is a tangent to parabola

$$y^2 = 4(x-1)$$

$$(mx)^2 = 4x - 4$$

$$m^2x^2 = 4x - 4$$

$$m^2x^2 - 4x + 4 = 0$$

tangent will occur when $\Delta = 0$

$$(-4)^2 - 4m^2 \cdot 4 = 0$$

$$16 - 16m^2 = 0$$

$$m^2 = 1$$

$$m = \pm 1$$

$\therefore y = x$ and $y = -x$ are tangents

$\therefore$ arguments lie between

$$-\frac{\pi}{4} < \arg z < \frac{\pi}{4}$$